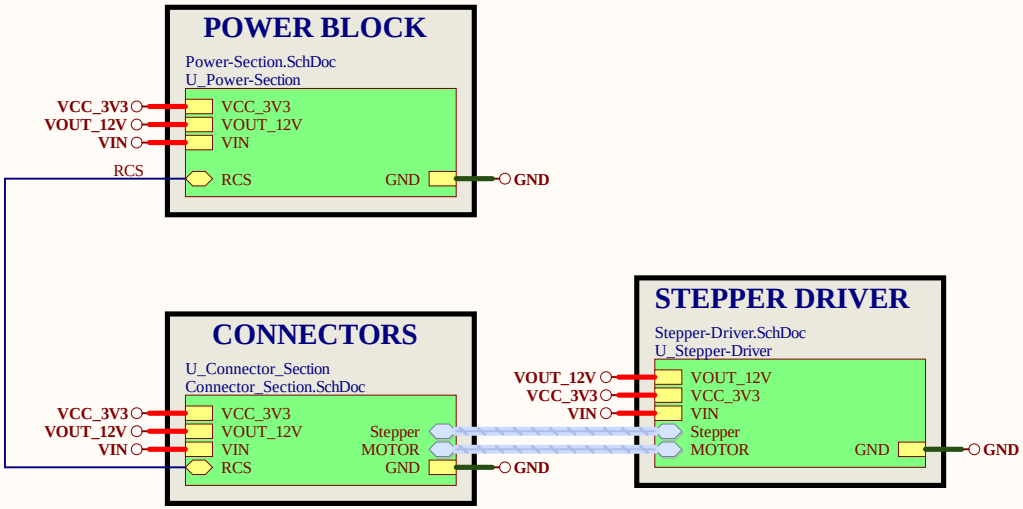
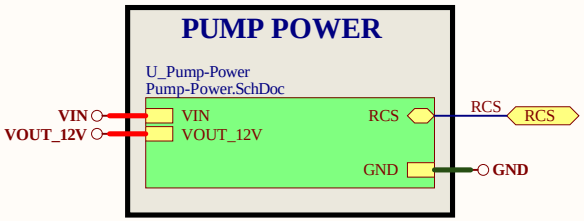
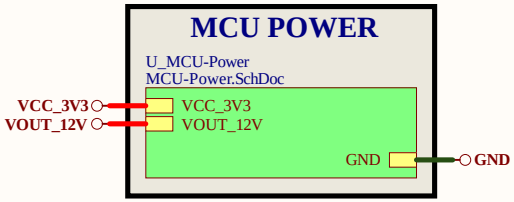
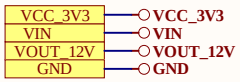
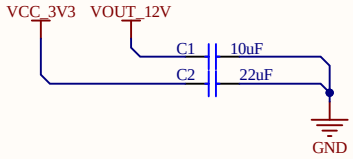

POWER BLOCK
Power-Section.SchDoc
U_Power-Section
VCC_3V3
VOUT_12V
VIN
RCS
VCC_3V3
VOUT_12V
VIN
RCS
GND
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
VCC_3V3
VOUT_12V
VIN
RCS
VCC_3V3
VOUT_12V
VIN
RCS
Stepper
MOTOR
GND
GND
STEPPER DRIVER
Stepper-Driver.SchDoc
U_Stepper-Driver
VOUT_12V
VCC_3V3
VIN
VOUT_12V
VCC_3V3
VIN
Stepper
MOTOR
GND
GND


PUMP POWER
U_Pump-Power
Pump-Power.SchDoc
VIN
VOUT_12V
VIN
VOUT_12V
RCS
GND
RCS
RCS
GND
GND

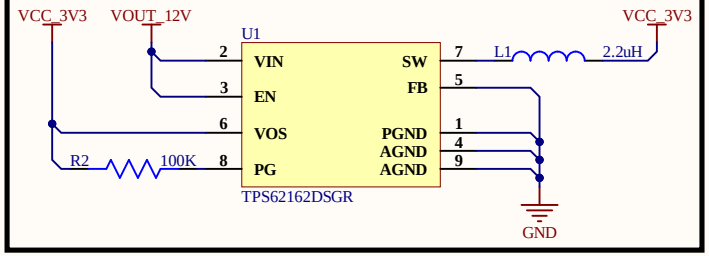

MCU POWER
U_MCU-Power
MCU-Power.SchDoc
VCC_3V3
VOUT_12V
VCC_3V3
VOUT_12V
GND
GND


VCC_3V3
VIN
VOUT_12V
GND
VCC_3V3
VIN
VOUT_12V
GND

DECOUPLING CAPACITORS

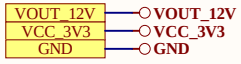


3V3 BUCK REGULATOR

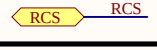


Fixed regulator.
VOUT= 3V3/1A

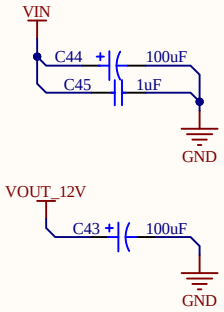
POWER SIGNALS



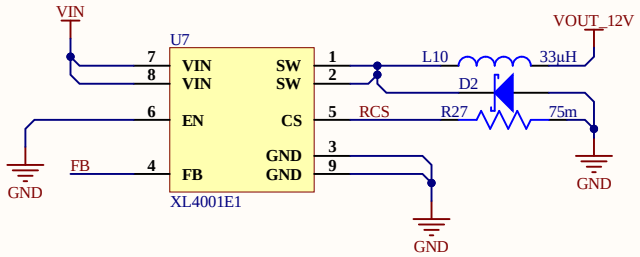
CURRENT SENSE INPUT



DECOUPLING CAPACITORS



12V BUCK REGULATOR

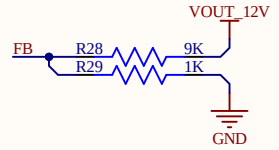


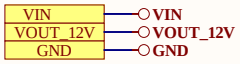
VIN=36V
VOUT=12V, where VOUT= 1.235(1+R2/R1)*
For 12V output, R2=9K and R1=1K
(Datasheet pg.3)

For IOUT=2A, RCS=75mA, (RCS=0.155/IOUT)

Reference: XL4001 Datasheet

FEEDBACK NETWORK

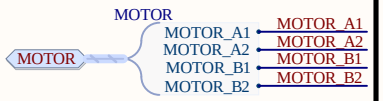




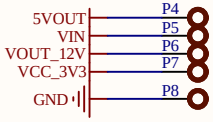
CONTROL SIGNALS



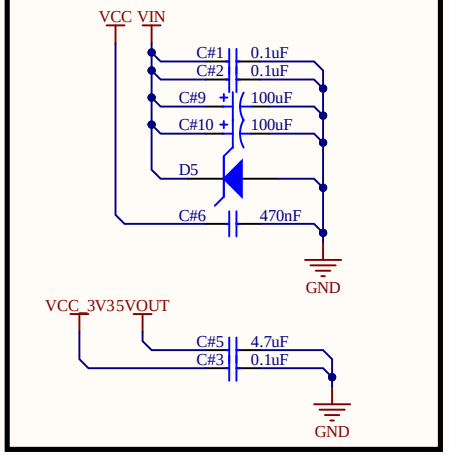
MOTOR OUT



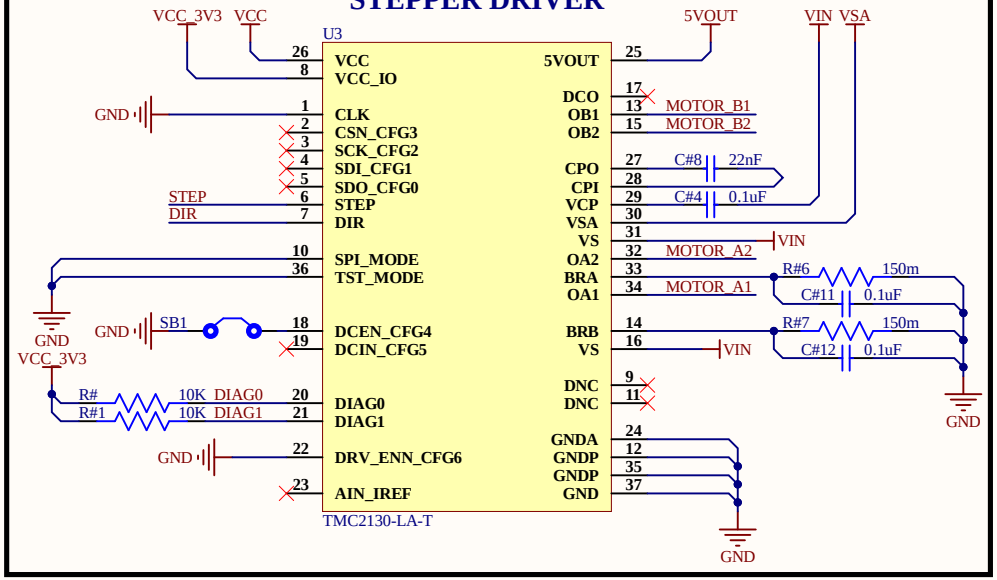
TEST POINTS



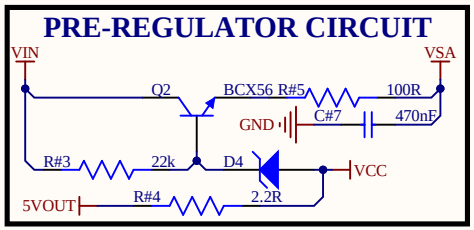
DECOUPLING CAPACITORS



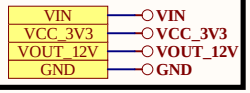
STEPPER DRIVER



PRE-REGULATOR CIRCUIT



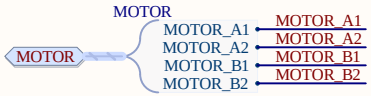
POWER SIGNALS



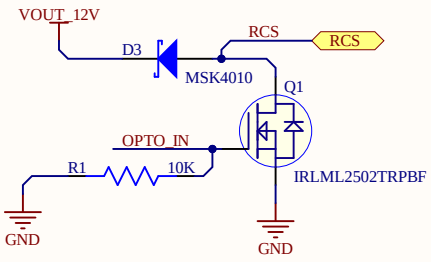
CONTROL SIGNALS



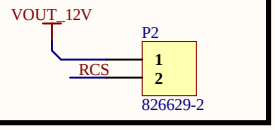
MOTOR OUT



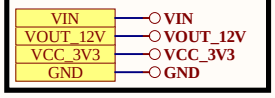
PUMP CONTROL



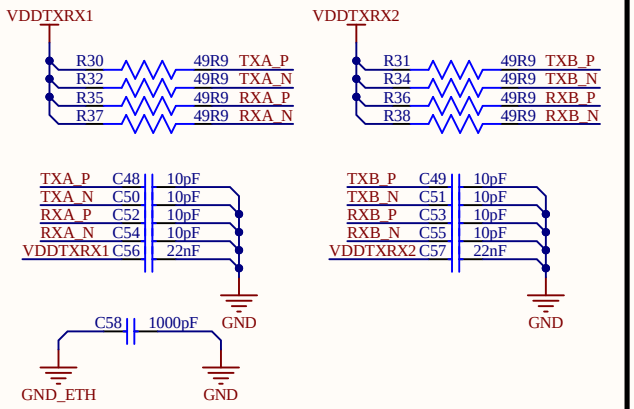
PUMP OUT



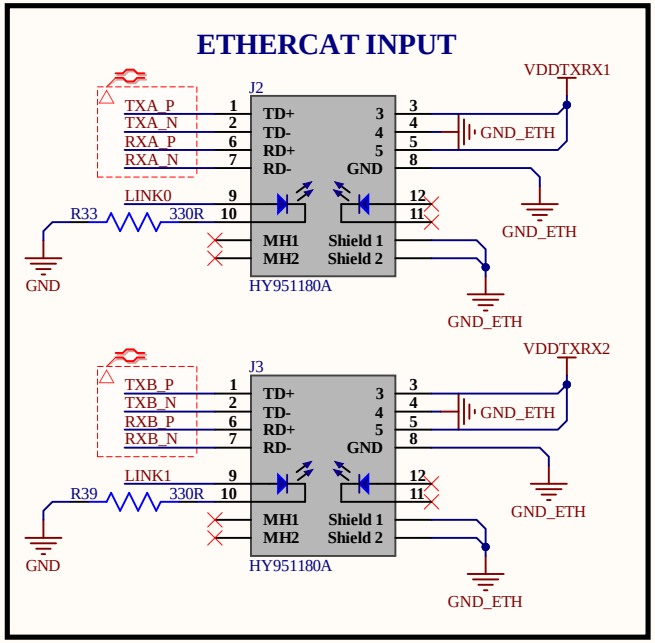
POWER SIGNALS



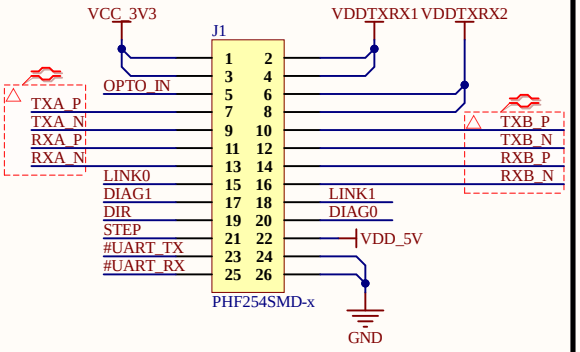
SUPPORTING CIRCUITRY



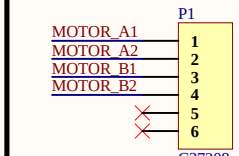
ETHERCAT INPUT



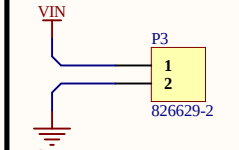
BOARD TO BOARD HEADER



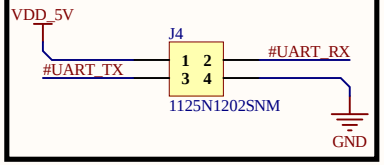
MOTOR OUT

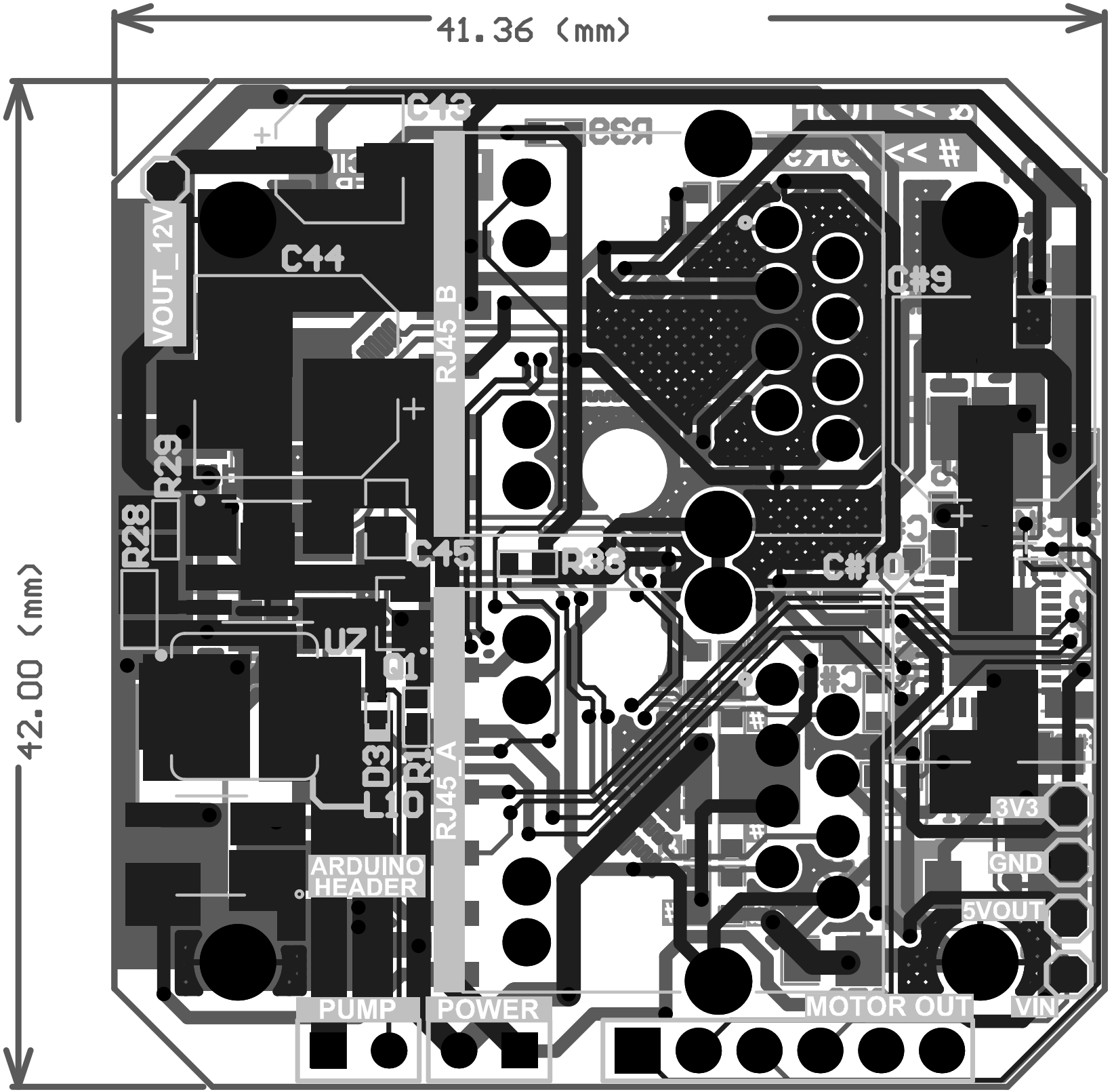


POWER IN



ARDUINO HEADERS




41.36 (mm)
42.00 (mm)
C43
C44
C45
C#9
C#10
R29
R28
R36
U2
D3
VOUT_12V
RJ45_B
RJ45_A
ARDUINO
HEADER
3V3
GND
5VOUT
PUMP
POWER
MOTOR OUT

Comment	Description	Designator	Footprint	LibRef	Quantity
10uF	CAP CER 10UF 25V X5R 0603	C1	Cap0603	GRM188R61E106MA73J	1
22uF	CAP CER 22UF 25V X5R 0805	C2	Cap0805	CC0805MKX5R8BB226	1
100uF	25V 5mm 10mm 100uF 100kHz SMD,5x10mm Aluminum Electrolytic Cap- SMD ROHS	C43	VKMB1001E101MV	VKMB1001E101MV	1
100uF	63V 10mm 8mm 100uF ±20% ,8x10mm Aluminum Electrolytic Cap	C44, C#9, C#10	VKMD1001J101MV	VKMD1001J101MV	3
1uF	CAP CER 1UF 100V X7S 0805	C45	Cap0805	GCM21BC72A105KE36L	1
10pF	CAP CER 10PF 50V C0G/NP0 0603	C48, C49, C50, C51, C52, C53, C54, C55	Cap0603	06035A100JAT2A	8
22nF	CAP CER 0.022UF 100V X7R 0603	C56, C57, C#8	Cap0603	CGA3E2X7R2A223K080AA	3
1000pF	1206 1 nF 2 kV ±10 % Tolerance X7R SMT Multilayer Ceramic Capacitor	C58	Cap1206	CC1206KKX7RDBB102	1
0.1uF	CAP CER 0.1UF 100V X7R 0603	C#1, C#2, C#3, C#4, C#11, C#12	Cap0603	CC0603KRX7R0BB104	6
4.7uF	CAP CER 4.7UF 25V X5R 0603	C#5	Cap0603	C1608X5R1E475M080AC	1
470nF	CAP CER 0.47UF 100V X7S 0805	C#6, C#7	Cap0805	C2012X7S2A474KT000N	2
PMEG3030EP,115	Diode Schottky 30V 3A 2-Pin SOD-128 T/R	D2	PMEG3030EP,115	PMEG3030EP,115	1
MSK4010	SOD-523 Schottky Barrier Diodes (SBD) ROHS	D3	MSK4010	MSK4010	1
MMSZ5V6CW	1μA@2V 40Ω Single 5.2V~6V 500mW 5.6V SOD123G Zener Diodes ROHS	D4	SOD123G	MMSZ5V6CW	1
SMBJ43A	8.6A 69.4V 47.8V 43V Unidirectional SMB(DO-214AA) TVS ROHS	D5	SMB	SMBJ43A	1
PHF254SMD-x	P=2.54mm Female Headers,SMD,ROHS	J1	PHF254SMD-x	PHF254SMD-x	1
HY951180A	Through Hole Ethernet Connectors/Modular Connectors (RJ45 RJ11) ROHS	J2, J3	HY951180A	HY951180A	2
1125N1202SNM	CONN HEADER 4 POS 2.54mm SMD	J4	1125N1202SNM	1125N1202SNM	1
2.2uH	IND SMD 1.5A 2.2uH ±20% 135mΩ	L1	MCS20FC-2R2MMP	MCS20FC-2R2MMP	1
33μH	FIXED IND 33UH 2.5A 189.2MOHM SM	L10	ETQ-P4M330YFN	ETQ-P4M330YFN	1
C37208	Straight 1 6 2.54mm Black Brass Plugin,P=2.54mm Pin Headers ROHS	P1	C37208	C37208	1
826629-2	CONN HEADER VERT 2POS 2.54MM	P2, P3	HDR1X2	826629-2	2
Test Point	Test Point ON	P4, P5, P6, P7, P8	TEST POINT	TEST POINT	5
IRLML2502TRPBF	MOSFET N-CH 20V 4.2A SOT23	Q1	IRLML2502TRPBF	IRLML2502TRPBF	1
BCX56	80V 500mW 1A 63@150mA,2V 130MHz 500mV@500mA,50mA NPN +150°C@(Tj) SOT-89(SOT-89-3) Bipolar Transistors - BJT ROHS	Q2	SOT-89	BCX56	1
10K	RES SMD 10K OHM 5% 1/10W 0603	R1, R#, R#1	Res0603	AC0603JR-0710KL	3
100K	RES SMD 100K OHM 5% 1/5W 0603	R2	Res0603	CRGH0603J100K	1
75m	RES 0.075 OHM 1% 2W 2512	R27	RES 2512	CSRN2512FK75L0	1
9K	RES SMD 9kΩ ±100ppm/°C 0.125W 0805	R28	RES 0805	0805W8F9001T5E	1
1K	RES SMD 1K OHM 1% 1/10W 0603	R29	Res0603	RC0603FR-071KL	1
49R9	RES 49.9 OHM 1% 1/10W 0603	R30, R31, R32, R34, R35, R36, R37, R38	Res0603	RC0603FR-0749R9L	8
330R	RES SMD 330R 100 mW 0603	R33, R39	Res0603	RC0603JR-07330RL	2
22k	RES SMD 22K OHM 0.1% 1/10W 0603	R#3	Res0603	RT0603BRE0722KL	1
2.2R	RES 2.2 OHM 1% 1/10W 0603	R#4	Res0603	RC0603FR-072R2L	1
100R	RES SMD 100 OHM 1% 1/10W 0603	R#5	Res0603	RC0603FR-07100RL	1
150m	RES SMD 150m ohms 1/4 1206	R#6, R#7	Res-1206	1206W4F150LT5E	2
Solder Bridge	Solder Bridge ON	SB1	Solder Bridge	Solder Bridge	1
TPS62162DSGR	Buck Switching Regulator IC Positive Fixed 3.3V 1 Output 1A 8-WFDFN Exposed Pad	U1	8-WSON (2x2)	TPS62162DSGR	1
TMC2130-LA-T	IC STEPPER MOTOR DVR SPI 36QFN	U3	TMC2130-LA-T	TMC2130-LA-T	1
XL4001E1	2A 40V SOIC-8_EP_150mil DC-DC Converters ROHS	U7	XL4001E1	XL4001E1	1